Statistical Foundations of AI

Conference Program

November 13–14, 2026 • UC Santa Barbara

Corwin Pavilion and University Center meeting rooms. All times are Pacific Standard Time.

Talks appear in presentation order. Each room's two-hour talk session is divided equally among its speakers. Panel discussions are highlighted.

Time and room	Friday November 13
Parallel Session 1	
1C Corwin 8:30–10:30 am **Statistical computation and structured modeling** Chair: Grace Yi	**Hani Doss** (University of Florida) Theoretical and Empirical Evaluation of a Grouped Gibbs Sampler for Parallel Computation in the Latent Dirichlet Allocation Model **Hao Helen Zhang** (University of Arizona) Dynamic Supervised Principal Component Analysis for High-Dimensional Classification **Yazhen Wang** (University of Wisconsin-Madison) Learning Relationship between Quantum Walks and Underdamped Langevin Dynamics **Ying Nian Wu** (UCLA) Quantum Field Theory as Generative Models, Emergent Gravity as Latent Variable Model **Bertrand Clarke** (U. Nebraska-Lincoln) Uncertainty Quantification via the Posterior Predictive Variance
1F Flying-A 8:30–10:30 am **AI agents and adaptive evaluation** Chair: Padhraic Smyth	**Jingshen Wang** (UC Berkeley) Agentifying AI Evaluation: What to Evaluate, When to Stop, and How to Estimate **Linjun Zhang** (Rutgers University) Statistical Evaluation of LLMs **David Talby** (John Snow Labs) Statistical design for medical AI safety benchmarks: precision targets, joint balance, and perturbation validity **Jason Klusowski** (Princeton University) Statistical Attribute Alignment for Black-Box Generative AI **Jie Ding** (University of Minnesota) The Future of AI Scientists: Emerging Directions and Fundamental Challenges **Ruoqing Zhu** (University of Illinois at Urbana-Champaign) Building AI Agents for Better Human-Agent Interaction
1H Harbor 8:30–10:30 am **Statistical learning for biomedical data** Chair: Yang Feng	**Jingjing Zou** (UC San Diego) Structured Self-Supervised Learning for Heterogeneous Wearable Data **Andrew Holbrook** (UCLA Biostatistics) Decoding Aging Neurons **Michael Daniels** (Department of Statistics, University of Florida) A case study of causal mediation using Bayesian nonparametrics and semiparametric corrections **Sam Behseta** (Kaiser Permanente Bernard J. Tyson School of Medicine) The Specification Gap: What Artificial Intelligence Leaves Unresolved in Statistical Inference **Lan Xue** (Oregon State University) Cluster Analysis of Longitudinal Associations Between Microbial Abundance and Environmental Factors **Sandra Safo** (University of Minnesota) Scalable Randomized Kernel Methods for Multiview Data Integration and Prediction
1S State Street 8:30–10:30 am **Generative models and learning dynamics** Chair: Jacob Bien	**Ning Ning** (Texas A&M University) Beyond Prediction: World Models for Understanding, Planning, and Action **Yuchen Zhou** (University of Illinois Urbana-Champaign) Optimal Convergence Analysis of DDPM for General Distributions **Shihao Yang** (Georgia Institute of Technology) Statistical Interpretation of AI Architecture and AI Agents for Statistical Research **Zhonghua Liu** (Columbia University) Gradient-Flow Optimization as Dynamic Random-Effects Inference with Applications to Deep Learning **Mingzhao Hu** (Mayo Clinic) ComposeCFM: Flow-Matching Synthetic Generation and Harmonization **Weining Shen** (UC Irvine) SportD: How do VLMs physically strategize?
10:30–11:00 am	**Break**
11:00 am–12:00 pm Corwin	**PLENARY TALK Susan Murphy (Harvard University)** Reinforcement Learning for Digital Health Interventions in the Dyadic Setting

Time and room	Friday November 13
12:00–1:00 pm	**Lunch**
Parallel Session 2	
2C Corwin 1:00–3:00 pm **Trustworthy AI and structured inference** Chair: Jin Zhou	**Weijie Su** (University of Pennsylvania) Alignment in Large Language Models: Statistical and Game-Theoretic Perspectives **Aaditya Ramdas** (Stanford University) Controlling the Probability of Unknown Unknowns in Pre-Deployment Testing of AI Systems **Padhraic Smyth** (UC Irvine) Prediction, Confidence, and Calibration in the Age of Deep Learning **Runze Li** (Penn State University) Federated Feature Selection with False Discovery Rate Control **Ji Zhu** (University of Michigan) Bias-Corrected Inference for Edge-Covariate Effects in Latent Space Network Models
2F Flying-A 1:00–3:00 pm **Regression and predictive inference** Chair: Yazhen Wang	**Oscar Hernan Madrid Padilla** (UCLA) Risk Bounds For Distributional Regression **Grace Yi** (University of Western Ontario) Residual Correction and Representation Learning for Robust Regression with Continuous Noisy Labels **Peijun Sang** (University of Waterloo) Model positive and unlabeled data with a generalized additive density ratio model **Matteo Sesia** (University of Southern California) Noise-Adaptive Conformal Classification with Marginal Coverage **Hongzhe Li** (University of Pennsylvania) Generation-Powered Inference for Distribution-Valued Outcomes **Min Xu** (Rutgers University) Estimation of the sub-Gaussian Parameter
2H Harbor 1:00–3:00 pm **Statistical learning and assisted inference** Chair: Ying Nian Wu	**Kun Chen** (University of Connecticut) Tree-Guided Feature Aggregation as Structured Representation Learning **Emma Jingfei Zhang** (Emory University) Beyond Point Predictions: Distribution-Informed Prediction-Powered Inference **Yang Feng** (New York University) Federated Transfer Learning with Differential Privacy **Kai Zhang** (UNC Chapel Hill) BELIEF in Dependence **Qiyang Han** (Rutgers University) Algorithmic inference via nonconvex gradient descent **Arun Kuchibhotla** (Carnegie Mellon University) Mean-Median gaps in real line and Inference in Infinite-dimensions
2S State Street 1:00–3:00 pm **Network models, representations, and experiments** Chair: Bertrand Clarke	**Xiwei Tang** (University of Texas at Dallas) HODGEDEPTH: SPATIAL REPRESENTATION LEARNING THROUGH GLOBAL GEOMETRY **Keith Levin** (University of Wisconsin—Madison) Testing for correlation between network structure and high-dimensional node covariates **Jacob Bien** (University of Southern California) Hierarchical Clustering With Confidence **Xiaoyue Maggie Niu** (Pennsylvania State University) A Dyadic Latent Space Model for Directed Networks **Jesús Arroyo** (Texas A&M University) Identifiability and Spectral Estimation in Multilayer Degree-Corrected Mixed Membership Models **Alexander Volfovsky** (Duke University) How to Experiment When You Don’t Yet Know What Matters
3:00–3:30 pm	**Break**
Parallel Session 3	
3C Corwin 3:30–5:30 pm **PANEL DISCUSSION** **Industry perspectives on trustworthy AI** Moderator: Hao Helen Zhang	**PANEL DISCUSSION Industry perspectives on trustworthy AI** **Moderator: Hao Helen Zhang** (University of Arizona) **David Talby** (John Snow Labs) **Sam Behseta** (Kaiser Permanente Bernard J. Tyson School of Medicine) **Guannan Zhai** (Amazon) **Yujia Deng** (Zoom Communications Inc.) **Haoda Fu** (Amgen) **Louis Ehwerhemuepha** (CHOC)

Time and room	Friday November 13
3F Flying-A 3:30–5:30 pm **Online learning and decision making** Chair: Lan Xue	**Xuan Bi** (University of Minnesota) Optimal Auditing of Sequential Decision Systems under Dynamical Constraints **Chengchun Shi** (London School of Economics and Political Science) Demystifying Group Relative Policy Optimization: Its Policy Gradient is a U-Statistic **Yuhua Zhu** (UCLA) Reinforcement Learning in the Physical World **Stefan Wager** (Stanford University) Estimating Dynamic Marginal Policy Effects under Sequential Unconfoundedness **Yufeng Liu** (University of Michigan) Structured Learning of Individualized Decision Rules with Complex Options
3H Harbor 3:30–5:30 pm **Statistical inference after learning** Chair: Hongzhe Li	**Lili Zheng** (University of Illinois Urbana - Champaign) LOCO-AdaMP: Built-in LOCO Inference for Adaptive Minipatch Ensembles with Enhanced Prediction **Armeen Taeb** (University of Washington) Consensus Tree Estimation with False Discovery Control via Partially Ordered Sets **Dan Kessler** (UNC Chapel Hill) Inference After Exploration: A Grammar for Adaptive Data Science **Tate Jacobson** (Oregon State University) Global Inference for High-Dimensional Quantile Regression with Heterogeneity **Qi Xu** (University of Minnesota) Towards Efficient Inference under Nonmonotone Missingness with General Imputation **Joshua Cape** (University of Wisconsin-Madison) Network Signflip Parallel Analysis for Selecting the Embedding Dimension
3S State Street 3:30–5:30 pm **Causal discovery and treatment effects** Chair: Hani Doss	**Yuqi Gu** (Columbia University) Discrete Causal Representation Learning **Carlos Cinelli** (University of Washington) Challenges in Causality: Sensitivity Analysis and Automation **Matteo Bonvini** (Rutgers University) On regression with estimated covariates and conditional effects given the propensity score **Yubai Yuan** (The Pennsylvania State University) Causal inference under interference with incomplete exposure mappings **Didong Li** (UNC Chapel Hill) When Can We Trust Synthetic Data for Statistical Inference? **Ying Zhou** (University of Connecticut) Learning from the Unseen: Offline Reinforcement Learning with Hidden Actions
5:30–6:30 pm	**Poster session** All posters share this session. See the poster directory. Venue to be confirmed.
7:00–8:30 pm Corwin	**BANQUET TALK Sudipto Banerjee (UCLA)** Carbs, Calories, and Computation: Serving Up Statistical Inference in the AI Ecosystem.

Time and room	Saturday November 14
Parallel Session 4	
4C Corwin 8:30–10:30 am **Biomedical modeling and AI-assisted inference** Chair: Hua Zhou	**Lily Wang** (George Mason University) Synthetic Surrogate Functional Regression for Neuroimaging Analysis **Lexin Li** (UC Berkeley) Integrating Statistics and AI for Neuroscience Research **Ying Guo** (Emory University) Task-Aware Statistical AI for Learning Directed Brain Connectomes from Neuroimaging Data **Fengzhu Sun** (University of Southern California) Statistical and AI Approaches for Investigating Viruses and Virus-host Interactions **Gang Li** (Department of Biostatistics, UCLA) SAPA: Safe and Adaptive Prediction Augmentation for Survival Analysis **Xinping Cui** (UC Riverside) Scalable Joint Modeling of Multiple Longitudinal Biomarkers and Competing Risks Time-to-Event Data: Applications to Mega-Scale Health Research

Time and room	Saturday November 14
4F Flying-A 8:30–10:30 am **Evaluation and inference with AI predictions** Chair: Veronica Berrocal	**Carey Priebe** (Johns Hopkins University) Performance guarantees for LLMs **Junwei Lu** (Harvard University) Language Model Preference Inference via Fisher Random Walk **Will Wei Sun** (Purdue University) Low Rank for Rank: Uncertainty-Aware Task-Specific LLM Ranking from Sparse Pairwise Comparisons **Boxiang Wang** (University of Iowa) ABC: Certified Selective Evaluation with a Panel of LLM Judges **Peng Wang** (University of Cincinnati) Building Statistical AI Tools for Comparative Analysis **Hongquan Xu** (UCLA) Prompt engineering via order-of-addition experiments and screening
4H Harbor 8:30–10:30 am **Structured representations and clustering** Chair: Ali Shojaie	**Shujie Ma** (UC Riverside) Learning What to Share: Adaptive Spectral Integration and Clustering of Multimodal Data **Mladen Kolar** (University of Southern California & MBZUAI) Borrowing Strength Across Studies: Structure-Aware Transfer Learning for High-Dimensional Models **Gongjun Xu** (University of Michigan) Beyond Vintage Rotation: Bias-Free Sparse Representation Learning with Oracle Inference **Jason Xu** (UCLA) Majorization-Minimization Algorithms in Clustering **Xiaohui Chen** (University of Southern California) GEM-KMeans: Memory-Efficient and Accurate Clustering on Massive Scale with GPU Optimization **Marianthi Markatou** (University at Buffalo) A study on kernels for discrete random variables
4S State Street 8:30–10:30 am **Reinforcement learning and adaptive decisions** Chair: Minge Xie	**Feifang Hu** (George Washington University) AI-Powered Adaptive Experiments: From Sequential Analysis to Intelligent Clinical Trials. **Lan Wang** (University of Miami) Quantile Temporal Difference Algorithm for Distributional Reinforcement Learning **Linglong Kong** (University of Alberta) Principled Fast and Meta Knowledge Learners for Continual Reinforcement Learning **Babak Shahbaba** (UCI) Uncertainty-Guided Control: Statistical Foundations for Brain-Inspired Decision-Making **Dehan Kong** (University of Toronto) Successive classification learning for estimating quantile optimal treatment regimes **Walter Dempsey** (University of Michigan) Future-Path Incremental Excursion Effects in Sequentially Randomized Trials
10:30–11:00 am	**Break**
11:00 am–12:00 pm Corwin	**PLENARY TALK Xihong Lin (Harvard University)** From Generative AI to Statistical Inference: Harnessing Synthetic Data for Validity and Power
12:00–1:00 pm	**Lunch**
Parallel Session 5	
5C Corwin 1:00–3:00 pm **PANEL DISCUSSION** **AI influence and the next generation of research** Moderator: Regina Liu	**PANEL DISCUSSION AI influence and the next generation of research** **Moderator: Regina Liu** (Rutgers University) **George Michailidis** (UCLA) **Bertrand Clarke** (U. Nebraska-Lincoln) **Heping Zhang** (Yale University) **Peter X. K. Song** (University of Michigan) **Radu Craiu** (University of Toronto) **Michael Daniels** (Department of Statistics, University of Florida) **Xiao Wang** (Purdue University) **Ying Wei** (Columbia University) **Michele Guindani** (UCLA)

Time and room	Saturday November 14
5F Flying-A 1:00–3:00 pm **Statistical learning and computational inference** Chair: Marianthi Markatou	**Minge Xie** (Rutgers University) Model-Free Inference for High-Dimensional Binary Classification Using Repro Samples **Faming Liang** (Purdue University) An Introduction to Stochastic Deep Learning **Weixin Yao** (UC Riverside) Online Kernel-Based Mode Learning **Ryan Giordano** (UC Berkeley) Black Box Variational Inference with a Deterministic Objective: Faster, More Accurate, and Even More Black Box **Debdeep Pati** (University of Wisconsin-Madison/Boston University) Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests **Ali Shojaie** (University of Washington) GLM Inference with AI-Generated Synthetic Data
5H Harbor 1:00–3:00 pm **Multi-view learning and data integration** Chair: Hongquan Xu	**Rong Ma** (Harvard University) Finding Shared Signals Across Many Datasets: Statistical Insights from RMT **Cong Ma** (University of Chicago) Bias-Corrected Subspace Intersection: Minimax-Optimal Shared Subspace Estimation in Multi-View Data **Tiffany Tang** (University of Notre Dame) Consensus dimension reduction via multi-view learning **Ziyi Liang** (UC Santa Barbara) Meta Fusion: A Unified Framework For Multimodal Fusion with Adaptive Mutual Learning **Xinzhou Guo** (HKUST) Stein-Encoder: A White-Box Supervised Encoder via Stein Identities in Multi-Modal Studies **Tian Gu** (Columbia University) Whom to trust and what to keep: adaptive Bayesian transfer learning from pretrained models
5S State Street 1:00–3:00 pm **Statistical learning with heterogeneous data** Chair: Hongtu Zhu	**Zhezhen Jin** (Department of Biostatistics, Columbia University) Subsampling for the analysis of large data **Gourab Mukherjee** (University of Southern California) Kernelized Stein Discrepancy for Learning from Heterogeneous Data **Jiwei Zhao** (University of Wisconsin-Madison) SADA: Safe and Adaptive Aggregation of Multiple Black-Box Predictions in Semi-Supervised Learning **Bei Jiang** (University of Alberta) fDA: A data-augmented predictive framework for generating fair synthetic labels with controlled fairness-faithfulness trade-offs **Lu Tang** (University of Pittsburgh) Fairness within Clinical Context: Stratified Tilted Risk Minimization for Medical Prediction **Veronica Berrocal** (UC Irvine) A Bayesian scalar-on-function kernel regression model for crop yield
3:00–3:30 pm	**Break**
Parallel Session 6	
6C Corwin 3:30–5:30 pm **Reliable learning and structured inference** Chair: Shujie Ma	**Hongtu Zhu** (UNC Chapel Hill) MLFM: Manifold-Based Latent Factor Discovery in Functional Brain Networks **Jian Kang** (University of Michigan) BROWNIE: Bayesian Random Weight Neural Networks as a Simple Alternative to Deep Learning **George Michailidis** (UCLA) Beyond Flat Minima: Sharpness-Aware Bilevel and Minimax Optimization. **Jinchi Lv** (University of Southern California) LIDS: LLM Summary Inference Under the Layered Lens **Qing Zhou** (Department of Statistics & Data Science, UCLA) Causal Discovery on Dependent Data
6F Flying-A 3:30–5:30 pm **Foundations of AI and distributed learning** Chair: Faming Liang	**Xiaoqian Liu** (UC Riverside) An Interpretable and Scalable Framework for Evaluating Large Language Models **Nabarun Deb** (University of Chicago Booth School of Business) Sample complexity bounds for categorical Markov random fields via Discrete Diffusions **Wenqing He** (University of Western Ontario) Beyond Marginal Coverage: Efficient Localized Conformal Prediction via Residual Rank Calibration **Rui Miao** (University of Texas at Dallas) TOPPO: Rethinking PPO for Multi-Task Reinforcement Learning with Critic Balancing **Jeffrey Zhang** (UC Riverside) Enhancing Federated Learning Methods using Third Order Optimization **Fei Xue** (Purdue University) Federated Learning for Feature Mismatch

Time and room	Saturday November 14
6H Harbor 3:30–5:30 pm **Learning from distributions and complex measurements** Chair: Lily Wang	**Linbo Wang** (University of Toronto) Estimating Wasserstein Barycenters from Sparse Deep Sequencing **Irina Gaynanova** (University of Michigan) Learning from multivariate distributions with nonparanormal transport **Hyunkeun Ryan Cho** (UCSD) Scalable four-level nested functional principal component analysis for large dense functional data **Soham Jana** (University of Notre Dame) Modeling with speckle noise in high-dimensional image sensing **Rui Duan** (Harvard T.H. Chan School of Public Health) Unsupervised Aggregation of Multiple Learning Algorithms **Wen Zhou** (New York University) A Latent Semiparametric Mixture Model for Robust Heterogeneity Learning in High Dimensions
6S State Street 3:30–5:30 pm **Learning from sequential and process data** Chair: Radu Craiu	**Xiucai Ding** (UC Davis) Transformers for Time Series: Statistical Optimality, Structural Exploration and Gradient Descent Consistency **Robert Lund** (UC Santa Cruz) Multiple Changepoint Detection in Non-Gaussian Time Series **Hernando Ombao** (Statistics, Bioengineering, and Applied Mathematics, KAUST) Dependence in Networks: Extremal Vulnerability and Granger Causality Beyond Linear Models **Chunming Zhang** (University of Wisconsin-Madison) Finite-Step Training of Deep ReLU Networks **Zhiliang Ying** (Columbia University) Process data and educational measurement **Yan Yu** (University of Cincinnati) (Mis)Understanding Benign Overfitting in Equity Return Prediction

Poster Session

Friday November 13 • 5:30–6:30 pm • Venue to be confirmed

All posters are presented in this single session. Listings are ordered by the first presenter's surname.

P01 Susmit Bhattacharyya (Colorado School of Mines) Marginal Distribution of the LatticeKrig coefficients under Spatial Autoregressive model with Spatial Extremes

P02 Stevenson Bolivar (California Polytechnic State University (Cal Poly)) Time-Varying Factor Models for Dynamic High-Dimensional Data

P03 Keivan Bolouri (UC Irvine) Aggregation Frequency and Active-Set Agreement in Federated Lasso

P04 Baiting Chen (UCLA) Common-agency games for multi-objective test-time alignment

P05 Xingran Chen (University of Michigan) Title to be confirmed *Topic: inference with machine learning imputations under general missingness.*

P06 Yifan Chen (UC Santa Barbara) Statistical Guarantees for Preference Learning with Convex Neural Networks

P07 Yassine Chouikh and Shannon Li (UC San Francisco) Verify-or-Abstain Evidence Pipelines: When Cryptographic Custody Is Necessary but Not Sufficient

P08 Chengyu Cui (Univ of Michigan - Ann Arbor, MI) Convexity in Disguise: A Theoretical Framework for Nonconvex Low-Rank Matrix Estimation

P09 Dongyu Cui (University of Minnesota) Observation-Level Watermarking and Detection for Tabular Data

P10 Ying Cui (Stanford University) Learning Interpretable Point-Based Clinical Risk Scores via Direct Optimization

P11 Annan Deng (New York University) Geometry-Aware Representation Learning for Functional Connectivity Matrices

P12 Prasanjit Dubey (Georgia Institute of Technology) Matching Rates and Optimal Allocation for Federated Probe-Logit Distillation under Heterogeneous Bandwidth Budgets

P13 Zijun Gao (USC Marshall) Adaptive sample splitting for randomization tests

P14 Daeyoung Ham (University of Texas at San Antonio) Black-Box Knowledge Transfer across Distinct Feature Sets

P15 Junchao Han (UC Santa Barbara) Off-Policy Evaluation for Linear Episodic Heterogeneous MDPs by Hierarchical Bellman Residual Minimization

P16 Rohan Hore (Stanford University) Conformal Changepoint Localization

P17 Pingbo Hu (Columbia University Irving Medical Center) Deep Learning for Continuous-Time Dynamic Survival Prediction With Time-Varying Covariates

P18 Yichen Hu (UC Davis) Adaptive Sparsity-Aware Projection-Assisted Spectral Embedding for High-Dimensional Noisy Data with Nonlinear Structures

P19 Yue Huang (University of Southern California) iLV-HD: Reconstructing microbial dynamics from high-dimensional compositional time series

P20 Hyeong Jin Hyun (University of Cincinnati) Transport-induced Approximate Bayesian Computation

P21 Ali Kaazempur-Mofrad (UCLA) Beyond a Single Answer: Proper Scoring of LLM Uncertainty

P22 Changwoo Lee (University of Florida) Transformers Can Learn Posterior Predictive Distributions In-Context

P23 Julianna Lee (Westat) Using Large Language Models to extract productivity measures from CVs for Program Evaluation

P24 Chunyin Lei (UC Santa Barbara) Risk-Aware Quantile Learning for Personalized Dynamic Treatment Regimes

P25 Dongyang Li (Department of Mathematics, UC Santa Barbara.) t-Muon: Tensorizing Muon with the t-SVD for Language Model Pretraining

P26 Pengtao Li (University of Southern California) Smoothed estimation of Wasserstein barycenters

P27 Xiaoke Li (Department of Statistics and Applied Probability, UC Santa Barbara) History-Aware Conformal Prediction for Irregularly Observed Longitudinal Data

P28 Ling Liang (The University of Tennessee, Knoxville) A Provably Convergent and Practical Algorithm for Gromov-Wasserstein Optimal Transport

P29 Qifeng Lin (UC Davis) Learning common structures from high-dimensional noisy biviews using cross-integral operator

P30 Yangyang Liu (Department of Quantitative and Computational Biology, University of Southern California, Los Angeles, CA, USA) MACONi: a marker- and composition-aware framework for meta-Hi-C contact normalization

P31 Jiecheng Lu (Georgia Institute of Technology) Rethinking Sequence Models: Autoregressive Alignment and Expressive Operator Design

P32 Muzhi Ma (UCLA) Mechanism Design for Heterogeneous LLM Advertising

P33 Ziling Ma (King Abdullah University of Science and Technology (KAUST)) Adaptive Multi-Scale Forecasting and Gate-Localized Conformal Prediction for Multivariate Nonstationary Time Series

P34 Elizabeth W. Miller (University of Virginia) Diagnostics for Individual-Level Prediction Instability in Clinical Prediction Models

P35 Simon Dovan Nguyen (University of Washington, Seattle) REALITrees: Rashomon Ensemble Active Learning for Interpretable Trees

P36 Ronan Perry (University of Washington) Inference after selection on randomized p-values

P37 Samya Praharaj (Rutgers University) Bandit Simulation for Average Reward Inference

P38 Chen Qian (UC Davis) Structural Classification of Locally Stationary Time Series Based on Second-order Characteristics

P39 Anik Roy (Department of Biostatistics and Bioinformatics, Emory University) Regression Analysis of Amplitude and Phase Components in Functional Data-Based Biomarkers

P40 Xinrui Ruan (UC Berkeley) Learning When to Seek Human Guidance for Best LLM Judge Identification

P41 Matias Shedden (University of Florida) Machine-learning-based subgroup identification and adaptive enrichment in a seamless Phase II/III trial

P42 Zhaiming Shen (Utah State University) Understanding In-Context Learning with Kernel Methods

P43 Zhongjie Shi (Georgia Institute of Technology) In-Context Learning Across Tasks with Transformers: Approximation and Generalization under Low-Dimensional Structure

P44 Shreyash Sinha (Department of Quantitative and Computational Biology, University of Southern California) Comparative exploration of gene-based and k-mer based features for predicting antimicrobial resistance

P45 Ao Sun (USC Marshall) LLM-Powered Predictive Inference with Online Text Time Series

P46 Pablo Suárez-Serrato (UC Santa Barbara) EuLearn: a 3D database for learning Euler characteristics

P47 Ye Tian (Penn State University) Semi-Supervised Representation Learning with Shared Factor Structure

P48 Yidan Tian (Michigan State University) Minimax-Optimal Transformer Classification for Functional Data with a Dense–Sparse Phase Transition

P49 Kaylee Vo (Harvard University) Variational Boosting for Physics-Informed Neural Networks

P50 Patrick Vossler (UC San Francisco) LLMs Judging LLMs: A Simplex Perspective

P51 Anela Leyan Wang (UCLA, Department of Biostatistics) Nonlinear subspace factor analysis via latent bottlenecks for integrating data across studies

P52 Kevin Wang (Univ of North Carolina At Chapel Hill) Can Generative Artificial Intelligence Survive Data Contamination? Theoretical Guarantees under Contaminated Recursive Training

P53 Kexin Wang (University of Southern California) DeepLINK-T: deep learning inference for time series data using knockoffs and LSTM with applications to metagenomic data

P54 Yizheng Wang (Hong Kong University of Science and Technology) Unsupervised Federated Multi-task Learning for Heterogeneous Tasks

P55 Joong-Ho Won (Seoul National University) Pareto Variational Autoencoder

P56 Runxiong Wu (University of Wisconsin-Madison) VarScope-MVAE: Learning Physically Scoped Variations in Multimodal Data

P57 Xiliu Wu (Indiana University Bloomington) Evaluating Tabular Foundation Models for Risk Prediction: A Benchmark against Traditional Machine Learning

P58 Qingyan Xiang (Department of Biostatistics, Vanderbilt University Medical Center) De-biased Machine Learning for Count Data: a Partially Linear Poisson Model Based on Neural Networks

P59 David Xie (Department of Statistics and Applied Probability, UC Santa Barbara) Analysis of Psychological Stress Response to Physiological Data from Wearable Devices: A Study on College Student Cohort.

P60 Tianmin Xie (University of Southern California) Conformal Inference for Open-Set and Imbalanced Classification

P61 Zhaoyu Xing (University of Notre Dame) SORHA: Semi-parametric Ordered Random Hypernetwork with Attributes

P62 Kathy Xu (UCSB PSTAT Department) Latent Restorative Patterns Linking Physical Activity to Acute Cardiovascular Protection in Wearable Health Data

P63 Minxing Xu (Northwestern University) Conflict in Medical Multi-Agent Reasoning: From Disagreement to Reliable Communication

P64 Wenlu Xu (UCLA) Physics-Informed Kernel Learning

P65 Haoran Yan (Georgia Institute of Technology) Neural Scaling Laws of Transformer Operator Network

P66 Mingsong Yan (UC Santa Barbara) In-Context Learning for Gaussian Kernel Ridge Regression with Transformers

P67 Tiantian Yang (University of Idaho) Directed Weighted Graph Neural Networks with Tensor Fusion for Multi-Omics Cancer Subtype Classification

P68 Yahong Yang (Binghamton University) Score Estimation and Approximation of the Diffusion Model

P69 Duoduo Danny Ying (Department of Statistics and Data Science, UCLA) Prompt engineering using order-of-addition experiments: An application to generating two-level fractional factorial designs

P70 Miaoqing Yu (UC Santa Barbara) BART for multi-outcome CATE estimation with Unmeasured Confounding

P71 Keming Zhang (Brown University) Interpretable and Uncertainty-Calibrated Bayesian Machine Learning for Survival Outcomes with Evolving Patient Histories

P72 Niuniu Zhang (UCLA Anderson School of Management) Joint Calibration of LLM Agents

P73 Sherry Zhang (University of Texas at Austin) Code availability is not enough: schemas for documenting and extending data analysis

P74 Yijiao Zhang (University of Pennsylvania) Causal Network Discovery from Interventional Count Data with Latent Linear DAGs

P75 Yikun Zhang (University of Chicago) Emputation: Identification-Guided Neural Imputation Framework

P76 Yuexia Zhang (The University of Texas at San Antonio) Extracting Causal Relational Rules for Medical Question-Answering Tasks using Large Language Models

P77 Shangcheng Zhao (Psychology & Brain Sciences, UCSB) Moral Shadowbanning: Is Conversational AI Less Helpful, Honest, or Harmless to Users Deemed to Have Poor Moral Character?

P78 Jialu Zhou (University of Maryland, College Park) Source-Fair Transport Calibration for Heterogeneous AI Evaluation

P79 Bicheng Zhu (Department of Statistics, UC Davis) Adaptive Clustering for Heterogeneous Multiplex Networks

P80 Jungang Zou (Department of Biostatistics at Columbia University) AI4BayesCode: From Natural Language Descriptions to Validated Modular Stateful Bayesian Samplers

Conference information: statfoundations-ai.github.io